Transcutaneous Spinal Stimulation & Gait Training Improves Functional Ambulation in Chronic TBI

Karen J. Nolan*, Member *IEEE,* Gregory R. Ames, Sharon Franco, Mitchell Tillman & Arun Jayaraman,

***Abstract*— Transcutaneous spinal cord stimulation (tSCS) combined with task specific gait training may improve functional mobility following traumatic brain injury (TBI). This study evaluated changes in functional mobility in a 20 year old female with chronic TBI (23 months post injury) following a 6 week treadmill based gait training with tSCS. Functional ambulation assessments included the Berg balance scale (BBS), functional gait assessment (FGA), timed up and go (TUG), 10 Meter Walk Test (10MWT), and 6 Minute Walk Test (6MWT) and were collected at baseline, short term and long term follow up. Improvements were demonstrated in balance, gait performance, mobility, walking speed, and endurance. These findings suggest that tSCS combined with gait training may support meaningful improvements in functional ambulation in individuals with chronic TBI.**

I. Introduction

Traumatic brain injury (TBI) affects 2.8 million people annually in the US and is a leading cause of long term disability [1]. TBI can disrupt sensorimotor control, resulting in gait and balance impairments that may persist during the chronic stages of recovery. This includes slower walking speed, decreased step length, and postural instability [2]. All of these pathological changes interfere with higher-level mobility skills necessary for full return to activities of daily living, full community participation, and vocational activities [3]. Activity-dependent neurorehabilitation remains the primary intervention for promoting long term motor recovery following TBI [3]. This approach is based on the principle of neuroplasticity, whereby repetitive, task-specific, goal-directed motor practice promotes reorganization of neural circuits and facilitates recovery of motor function [4]. Recent advances in rehabilitation have evaluated the combination of rehabilitative training with neuromodulation of the brain and/or the spinal cord as means to enhance the excitability of motor circuits and to increase training efficacy promoting motor recovery [5,6]. The proposed investigation will evaluate the efficacy and safety of transcutaneous spinal stimulation (tSCS) to promote functional motor recovery for individuals with chronic TBI. The fundamental hypothesis guiding this proposal is that transcutaneous spinal cord stimulation (tSCS) combined with gait training (WALK) for individuals with chronic TBI will lead to improved functional ambulation.

II. Methods

A. Participants

A single participant (20 year old female) diagnosed with TBI (23 months post injury, height 1.63 m, 54.43 kg) enrolled in the study. All procedures were conducted in accordance with the institutional review board and informed consent was obtained prior to participation.

Participants completed a 6 week (3 times per week, 18 sessions) treadmill based gait training program (WALK) focused on improving symmetry, kinematic quality and walking intensity. Transcutaneous spinal cord stimulation (tSCS) was delivered using a portable stimulator (Digitimer DS8R®, USA). Biphasic stimulation was applied over the thoracolumbar region using a cathode electrode (7.5 X 13cm) positioned between T11/T12 and L1/L2 vertebrae (corresponds to spinal networks that innervate leg muscles) and two anodes placed bilaterally over the iliac crests (as shown in Fig. 1). During treadmill based gait training stimulation was delivered at 30 Hz base frequency, a 5 kHz carrier frequency and a 100 µs pulse width.

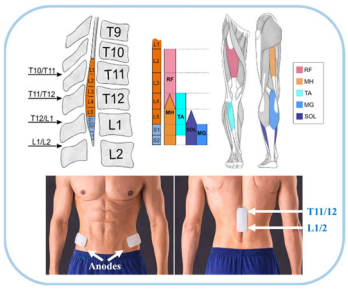

Fig. 1: tSCS electrode placement.

B. Data Collection Procedures

Functional ambulation assessments were collected at three timepoints, baseline, post intervention and long term follow up. Outcome measures:

- *Berg balance scale* (BBS) – a task performance test of static and dynamic balance consisting of 14 items of increasing difficulty graded on a five-point ordinal scale of 0 to 4 (maximum score is 56).
- *Functional Gait Assessment (FGA)* – a measure of dynamic

*Research supported by the DOW, CDMRP HT9425-24-1-1014 (MPI KJ Nolan, A Jayaraman).

K. J. Nolan is with the Kessler Foundation, West Orange, NJ, & Rutgers–New Jersey Medical School, Newark, NJ USA (corresponding author, phone: 973-324-3544; email: knolan@kesslerfoundation.org).
G. R. Ames is with the Kessler Foundation, West Orange, NJ 07052 USA (email: games@kesslerfoundation.org); C. M. Dandola is with the Kessler Foundation, West Orange, NJ 07052 USA (email: cdandola@kesslerfoundation.org); S. Franco is with the Kessler Foundation, West Orange, NJ 07052 USA (email: sfranco@kesslerfoundation.org);
M. Tillman is with Shirley Ryan AbilityLab, Chicago, IL 60611 USA (email: mtillman@sralab.org; and A. Jayaraman are with Shirley Ryan AbilityLab, Chicago, IL 60611 USA (email: ajayaraman@sralab.org);

balance during walking, consisting of 10 tasks, each item is scored from 0 to 3 with a maximum score of 30.

- *Timed up and go* (TUG) – a measure of balance and walking performance; participants rise from a chair, perform a 3-meter walk, and return to a seated position. The time to successfully complete the task is recorded (s).
- 10MWT – a performance measure of functional mobility, provides a measure of walking speed in meters/second.
- *6MWT* – an endurance measure where the total walking distance is recorded.

III. Results

Transcutaneous spinal cord stimulation (tSCS) combined with gait training resulted in measurable improvements in functional mobility post intervention and long term follow up (Table 1, Fig. 2). All intervention sessions were completed, and stimulation was tolerated during treadmill gait training with no adverse events. Balance improved as indicated by increases in BBS from 48 at baseline to 53 post intervention and 56 at long term follow up. FGA scores improved from 22 to 26, indicating enhanced dynamic balance and gait performance. TUG time changed from 10.62 s at baseline to 11.41 s post intervention and improved to 8.21 s at long term follow up. Walking speed increased from 1.14 m/s at baseline to 1.49 m/s post intervention and 1.42 m/s at long term follow up. Walking endurance also improved, with 6MWT distance increasing from 500.9 m at baseline, 522.0 m post intervention and 528.5 m at long term follow up.

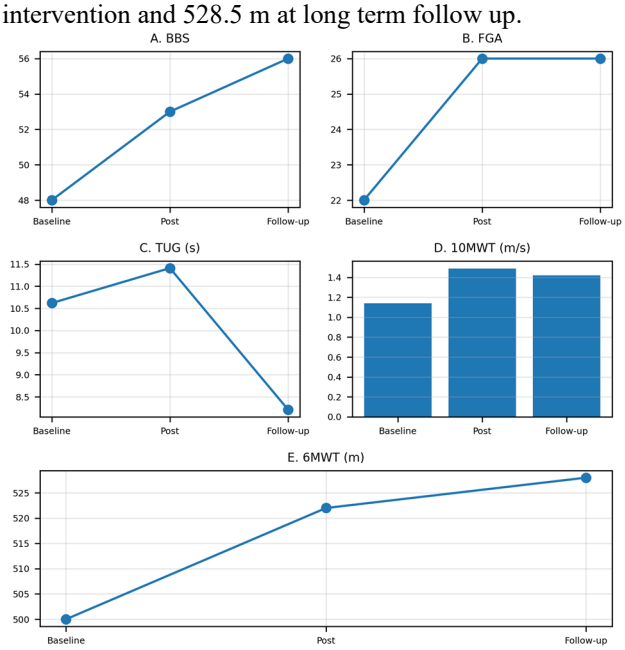

Fig. 2. Functional outcomes across timepoints. (A) Berg Balance Scale (BBS), (B) Functional Gait Assessment (FGA), (C) Timed Up and Go (TUG), (D) 10-Meter Walk Test (10MWT) speed, and (E) 6-Minute Walk Test (6MWT).

IV. Discussion

These preliminary data demonstrated that tSCS combined with gait training may improve functional mobility, balance and endurance in a chronic TBI participant. The participant improved balance post intervention (BBS) which may indicate a decrease in fall risk. Post training the individual also improved their ability to modify balance while walking in the presence of external demands (FGA). The participant met the MCID criteria for a meaningful clinical change in FGA score of 4.2 points. The TUG score correlates with balance, gait speed, and functional capacity and is widely used as a measure of physical mobility. The participant improved their performance on the TUG after 6 weeks of training and met the minimal detectable change (MDC) of 2.9 seconds. Changes in walking speed also reached the MCID of 0.06 m/s indicating a meaningful change in gait speed.

Improvements in BBS, FGA and TUG suggest meaningful gains in dynamic balance and mobility, while changes in walking speed and 6MWT indicate improved functional ambulation and endurance. These findings suggest that neuromodulation combined with activity based rehabilitation may support improvements in functional ambulation. While neurophysiological mechanisms were not the primary focus of this analysis, improvements in functional performance are consistent with motor system responsiveness.

More research is needed in a larger sample with individuals with varying levels of impairment to fully understand how tSCS combined with gait training can impact overall recovery. These improvements in mobility suggest that tSCS combined with gait training may promote meaningful functional changes in motor recovery.

V. Conclusion

tSCS combined with treadmill based gait training improved functional ambulation in a chronic TBI participant. Improvements across multiple clinical measures suggest meaningful gains in functional ambulation. This preliminary data demonstrates the potential of tSCS as an adjunct in rehabilitation for individuals with TBI. Future research will expand this preliminary data and continue to understand the potential of tSCS combined with gait training in chronic TBI.

References

[1] Taylor, C.A., et al., (2017). Traumatic brain injury–related emergency department visits, hospitalizations, and deaths—United States, 2007–2013, MMWR Surveillance Summaries, 66(9), 1–16.

[2] Dever, A., et al. (2022). Gait Impairment in Traumatic Brain Injury: A Systematic Review. Sensors, 22(4).

[3] Marshall, S.J., et al., (2022) Gait impairments in traumatic brain injury: A systematic review, Sensors, 22(4).

[4] Kleim, J.A., & Jones, T.A.. (2008) Principles of experience-dependent neural plasticity: Implications for rehabilitation after brain damage, J. Speech Lang. Hear. Res., 51(1) S225–S239.

[5] Clayton, E., et al. (2016). Brain stimulation: Neuromodulation as a potential treatment for motor recovery following traumatic brain injury. Brain Res, 1640(Pt A), 130-138.

[6] Hofer, A.S., and Schwab, M.E., (2019) Enhancing rehabilitation and functional recovery after brain and spinal cord trauma with electrical neuromodulation, Current Opinion in Neurology 32(6), 828–835,.

[7] Skiadopoulos A, & Knikou M (2025) Optimal sigmoid function models for analysis of transspinal evoked potential recruitment curves recorded from different muscles. PLOS ONE 20(1)

[8] Goetz SM, et al. (2014) A novel model incorporating two variability sources for describing motor evoked potentials. Brain Stimul. Jul-Aug;7(4):541-52.

[9] Tyagi, V. et al. (2025) Hierarchical Bayesian estimation of motor-evoked potential recruitment curves yields accurate and robust estimates. Brian Stimul. Nov-Dec; 18(6): 1855-1870